

MICROSATELLITE PRIMERS FOR THE ENDANGERED BEECH TREE, *FAGUS HAYATAE* (FAGACEAE)¹

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- **Premise of the study:** Microsatellite primers were developed for the endangered species *Fagus hayatae* (Fagaceae) to investigate the genetic diversity of the population and to investigate species delimitation within *Fagus*.
- **Methods and Results:** Ten polymorphic simple sequence repeat markers were developed for *F. hayatae* using a magnetic bead enrichment method. The primers amplified trinucleotides, hexanucleotides, and complex repeats with six to 16 alleles per locus. The observed and expected heterozygosities across loci varied, with a range of 0.05–0.71 and 0.63–0.91, respectively. Most of the primers also amplified DNA from *F. crenata*, *F. grandifolia*, *F. japonica*, *F. longipetiolata*, *F. lucida*, *F. orientalis*, and *F. sylvatica*.
- **Conclusions:** These results indicate that these polymorphic loci of *F. hayatae* will be potentially useful for future studies of the population genetic diversity within the species. In addition, the interspecific amplification indicates that these transferable microsatellite markers will also be useful for future phylogeographic and speciation studies among close *Fagus* species.

Key words: beech tree; Fagaceae; *Fagus hayatae*; genetic diversity; magnetic bead enrichment method; microsatellites; simple sequence repeat markers.

Fagus hayatae Palib. ex Hayata (Fagaceae) is an endangered beech tree, with only a few populations isolated in disjunct areas of southern China and northern Taiwan. This species was categorized at the vulnerable level in the IUCN Red List of Threatened Species. Individuals from each area are slightly different morphologically and have been treated as taxonomically distinct, including *F. pashanica* C. C. Yang in southwestern China and *F. hayatae* var. *zhejiangensis* M. C. Liu & M. H. Wu ex Y. T. Chang & C. C. Huang in southeastern China (Huang et al., 1999). In Taiwan, *F. hayatae* is an endangered species with two remaining populations. The tree grows in deciduous forests on Mt. Chatian and Mt. Tau, and hundreds of individuals have been found (Liao, 1996). However, no genetic diversity studies have been conducted. Although several microsatellite markers from related beech taxa have been developed (Pastorelli et al., 2003; Asuka et al., 2004; Vornam et al., 2004), the microsatellite primers were found to be not applicable in *F. hayatae* by a preliminary test. Thus, we developed 10 polymorphic microsatellite markers to investigate the genetic diversity of *F. hayatae*.

The transferability of the microsatellite primers developed in this study was also tested in other species, specifically, the native populations of *F. crenata* Blume from Japan, *F. longipetiolata* Seemen from China, and *F. sylvatica* L. from England. We also tested whether the sequences targeted beech taxa,

including *F. grandifolia* Ehrh., *F. lucida* Rehder & E. H. Wilson, *F. orientalis* Lipsky, and *F. japonica* Maxim.

METHODS AND RESULTS

Material from four *Fagus* species were collected from one, two, or three populations in their native habitats (Appendix 1). For *F. hayatae*, material was collected from the two remaining populations in Taiwan. To test the transferability of the microsatellite loci, we collected material from four other taxa in botanical gardens, including *F. grandifolia*, *F. lucida*, *F. orientalis*, and *F. japonica* (Appendix 2). The genomic DNA from the dried leaves of each species was extracted using a modified cetyltrimethylammonium bromide (CTAB) method. The selective capture of microsatellite loci using modified amplified fragment length polymorphism and magnetic beads was used as a methodological strategy to clone the microsatellite loci. The microsatellite isolation procedure followed the method of Chiang et al. (2011), as modified by Zane et al. (2002). Fragments of the genomic DNA from one individual were digested into segments by the restriction enzyme *MseI* (Promega Corporation, Madison, Wisconsin, USA), purified, and ligated to the double-stranded adapter (complementary oligo A: 5'-TACTCAGGACTCAT-3'; phosphorylated oligo B: 5'-GACGATGAGTCCTGAG-3'). The *MseI*-digested library was enriched using PCR with 20 ng template DNA, 10 pmol adapter-specific primers (5'-GATGAGTCCTGAGTAAN-3'), 2 µL 10× reaction buffer, 2 mM dNTP mix, 2 mM MgCl₂, 0.5 U *Taq* DNA polymerase (Promega Corporation), and sterile water to 20 µL in total. PCR products were denatured and hybridized to three 5'-biotinylated oligonucleotide probes (AG)₁₅, (AC)₁₅, and (GT)₁₀ in 200 µL hybridization solution at 68°C for 1 h, and 1 mg of Streptavidin Magnetic Particles (Promega Corporation) was mixed to the PCR products to capture the hybridizations at 42°C for 2 h. The enriched (AG)₁₅, (AC)₁₅, and (GT)₁₀ fragments were washed in high- and low-salt solutions. Purified enriched fragments were used as templates for 25 cycles of PCR amplification using *MseI*-N, and then amplicons were cloned using pGEM-T Easy Vector System (Promega Corporation) and screened using PCR (primer pairs: (AG)₁₀, (AC)₁₀, or (GT)₇ and SP6 or T7). In total, 142 plasmids from selected clones were purified and subsequently sequenced using an ABI 3730 sequencer (Applied Biosystems, Foster City, California, USA). A total of 36 sequences displayed perfect repeats, as detected by Tandem Repeats Finder version 4.04

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TABLE 1. Characteristics of 10 polymorphic microsatellite loci from *Fagus hayatae* (Fagaceae).

Locus	Primer sequences (5'–3')	Repeat motif	Allele size (bp)	T_a (°C)								GenBank accession no.
				hay	cre	lon	syl	jap	luc	ori	gra	
TB-100	F: ATGGTTGAAGAGGTCCATTG R: AAGTCATCCACGCCACAGTC	(TTC) ₉ N(TGT) ₃ (TC) ₁₀	431–461	64	64	64	64	64	64	64	64	HE687310
TB-101	F: TGCTTGGTTTTCGAGAAGC R: AACAGTGGCTTGCACAAAC	(AGT) ₂ (TGT) ₂₀	195–216	64	62	62	62	64	62	64	64	HE687311
TB-129	F: CTAGGAGCAGCACCAGCAG R: CTGTTGCTATTCCTTTTCGTG	(GT) ₈ (GA) ₁₂	220–280	62	62	62	62	62	62	62	62	HE687312
TB-133	F: TCGCGATGATCCGACTCCGAAC R: CCATCTCCAGTTCAAGACCCA	(TGTGCATGTG) ₃ (TG) ₄	271–313	62	62	62	62	62	—	62	62	HE687313
TB-140	F: TCCAAACTTGCTGCTGTTCC R: CAAAAGTGACATACGTTACG	(TGT) ₂₂	207–249	64	63	63	63	64	63	64	—	HE687314
TB-142	F: GGCATGCTTGGAACCCAG R: CCTTAGGTCCTTCTCACCGA	(GTT) ₈	270–360	64	64	64	64	64	64	64	64	HE687315
TBN-18	F: TTAGAAACCCACGCGTGC R: CTCTATCGGTGACAAACATCG	(CA) ₈ (CT) ₁₂	212–268	60	60	60	—	60	60	—	—	HE687316
TBN-24	F: GATGGTGATAACTGAGTGAG R: CTAGCTTGTGCTCTATGC	(CATTCATATT) ₂ (GA) ₂₃	226–258	60	60	60	60	60	60	60	60	HE687317
TBN-50	F: ATGAGGGTTTGGAAATGCT R: TCTACTATGATCCGAGGT	(TTG) ₇	198–303	54	54	54	54	54	54	54	54	HE687318
TBN-61	F: GGTGGAGTACCAGAGCTGCA R: TGTCAATAGGCACAGGTTTCG	(TGTGG) ₁₅	280–380	60	60	60	60	60	60	60	60	HE687319

Note: — = not applicable; *cre* = *F. crenata*; *gra* = *F. grandifolia*; *hay* = *F. hayatae*; *jap* = *F. japonica*; *lon* = *F. longipetiolata*; *luc* = *F. lucida*; *ori* = *F. orientalis*; *syl* = *F. sylvatica*; T_a = optimized annealing temperature.

(Benson, 1999), and primer pairs in flanking regions for each microsatellite locus were designed using FastPCR software version 5.4 (Kalendar et al., 2009). The parameters for the specific primer design were a PCR product size range from 200 to 450 bp, an optimum annealing temperature of 55°C, and a GC content ranging from 40% to 60%. The optimal annealing temperature was determined by a gradient PCR protocol using the Labnet MultiGene 96-well Gradient Thermal Cycler (Labnet, Edison, New Jersey, USA) for an individual of each beech species, and polymorphisms were evaluated using 21 and 20 individuals of *F. hayatae* from the two remaining populations in Taiwan. Lastly, 10 of 36 microsatellite loci with perfect-repeat target PCR products and polymorphic results were chosen for further study (Table 1). To evaluate the inter-specific transferability, three species from native populations (*F. crenata*, *F. longipetiolata*, *F. sylvatica*) and four species from botanical gardens (*F. grandifolia*, *F. lucida*, *F. orientalis*, *F. japonica*) were examined using the optimal annealing temperature (Table 1) to amplify the DNA and 10% PAGE electrophoresis using a 25- or 50-bp DNA Step Ladder (Promega Corporation) to separate and evaluate the allele sizes.

All 10 new microsatellite loci identified in *F. hayatae* were successfully amplified and polymorphic in the four species from native populations, except primer TBN-18 for *F. sylvatica* (Table 2). Other untested primer pairs are listed in Appendix 3. Using Arlequin 3.11 (Excoffier et al., 2005), the number of alleles

per locus (A), number of effective alleles (A_e), observed heterozygosity (H_o), and expected heterozygosity (H_e) were calculated. As shown in Table 2, A ranged from six to 14 and six to 16 in two populations of *F. hayatae* and from five to 24, three to 17, and five to 19 in *F. crenata*, *F. longipetiolata*, and *F. sylvatica*, respectively. A_e varied from 2.69 to 11.31 and 2.53 to 11.59 in two populations of *F. hayatae* and from 3.68 to 20.48, 2.97 to 13.07, and 4.25 to 14.52 in the other three species. H_o and H_e ranged from 0.05 to 0.71 and 0.61 to 0.91 in *F. hayatae*, 0.19 to 0.81 and 0.73 to 0.95 in *F. crenata*, 0.05 to 0.79 and 0.65 to 0.92 in *F. longipetiolata*, and 0.07 to 0.76 and 0.76 to 0.93 in *F. sylvatica*, respectively (Table 2). Significant deviations in Hardy–Weinberg equilibrium (HWE) (Table 2) were detected in six, seven, four, and five of the 10 microsatellite loci in the four beech taxa. No significant linkage disequilibrium was discovered between the pairs of loci.

To test the transferability of these microsatellite loci, we tested these primers in other species from botanical gardens, including *F. grandifolia*, *F. lucida*, *F. orientalis*, and *F. japonica*. One to four samples of each species were used in the evaluation of cross-amplification and almost all loci were transferable (Table 3). Nevertheless, the phylogeographic patterns among related beech species in Asia remain unclear. The primer set of these microsatellite markers with high transferability represents a useful set of genetic markers for interspecific research on such topics as phylogeography.

TABLE 2. Average genetic diversity for four species of *Fagus* using the 10 newly developed polymorphic microsatellite markers.

Locus	<i>F. hayatae</i>																			
	Mt. Chatian ($N = 21$)				Mt. Tun ($N = 20$)				<i>F. crenata</i> ($N = 16$)				<i>F. longipetiolata</i> ($N = 14$)				<i>F. sylvatica</i> ($N = 15$)			
	A	A_e	H_o	H_e	A	A_e	H_o	H_e	A	A_e	H_o	H_e	A	A_e	H_o	H_e	A	A_e	H_o	H_e
TB-100	9	6.26	0.48	0.84*	12	10.67	0.05	0.91*	7	4.38	0.31	0.77*	3	2.97	0.71	0.66	9	6.16	0.07	0.84*
TB-101	6	4.45	0.57	0.78	7	3.76	0.65	0.73	6	5.07	0.44	0.80*	7	5.94	0.57	0.83	5	4.25	0.47	0.76
TB-129	7	4.69	0.57	0.79	7	2.92	0.50	0.66	8	4.41	0.69	0.77	4	2.88	0.23	0.65*	7	4.95	0.67	0.80
TB-133	8	4.03	0.52	0.75	14	8.33	0.25	0.88*	11	7.53	0.81	0.87*	10	8.91	0.17	0.89*	8	5.84	0.73	0.83*
TB-140	11	7.60	0.67	0.87*	14	11.11	0.45	0.91*	13	9.66	0.44	0.90*	9	7.00	0.05	0.86*	13	7.76	0.60	0.87*
TB-142	14	9.38	0.48	0.89*	9	6.02	0.05	0.83*	14	9.48	0.38	0.89*	17	13.07	0.71	0.92	11	7.76	0.73	0.87
TBN-18	13	11.31	0.29	0.91*	16	11.59	0.30	0.91*	5	3.68	0.38	0.73*	11	7.69	0.21	0.87*	—	—	—	—
TBN-24	6	2.69	0.67	0.63	6	2.53	0.70	0.61	24	20.48	0.81	0.95	9	4.90	0.79	0.80	19	14.52	0.76	0.93
TBN-50	12	8.40	0.52	0.88*	13	8.25	0.45	0.88*	14	10.45	0.19	0.90*	5	4.51	0.64	0.78	11	6.82	0.13	0.85*
TBN-61	10	5.76	0.71	0.83	10	5.44	0.70	0.82	5	3.74	0.50	0.73	6	3.56	0.79	0.72	8	5.29	0.53	0.81*

Note: — = unamplified loci; A = number of alleles; A_e = number of effective alleles; H_e = expected heterozygosity; H_o = observed heterozygosity; N = sample size.

*Deviation from Hardy–Weinberg equilibrium: $P < 0.05$.

TABLE 3. Applicability of microsatellite loci in four *Fagus* species.^a

Locus	<i>F. japonica</i> (<i>N</i> = 4)	<i>F. lucida</i> (<i>N</i> = 1)	<i>F. orientalis</i> (<i>N</i> = 1)	<i>F. grandifolia</i> (<i>N</i> = 2)
TB-100	3	1	1	2
TB-101	3	1	1	1
TB-129	3	2	2	3
TB-133	2	—	2	2
TB-140	5	1	2	—
TB-142	5	1	1	2
TBN-18	3	1	—	—
TBN-24	4	1	1	3
TBN-50	3	2	2	2
TBN-61	3	2	1	1

^aFor loci that amplified, the number of alleles is given; unamplified loci are denoted with —.

CONCLUSIONS

Ten polymorphic microsatellite loci identified in *F. hayatae* were characterized for the two remaining populations of *F. hayatae* in Taiwan and for three related species (*F. crenata*, *F. longipetiolata*, and *F. sylvatica*). In addition, successful amplification in four other beech taxa indicated the high transferability of these primer pairs. For conservation efforts, these markers will be useful for future studies on the estimation of the genetic variability of endangered *F. hayatae* in Taiwan. At the interspecies level, the set of markers represents a potential tool to reconstruct phylogeographic patterns in related *Fagus* taxa.

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APPENDIX 1. Information on voucher specimens for four *Fagus* species collected in the field. All specimens are deposited at the Taiwan Forestry Research Institute (TAIF).

Species	Locality	<i>N</i>	Latitude (°N)	Longitude (°E)	Voucher no.
<i>F. hayatae</i>	Mt. Chatian Nature Reserve, Taiwan	21	24°47'30.77"	121°27'09.34"	Cheng8314
	Mt. Tun, Taiwan	20	30°34'34.21"	121°36'32.04"	Ju 17
<i>F. crenata</i>	Mei Prefecture, Japan	5	34°38'46.7"	136°23'12. 3"	Ju 135
	Chubu Sangaku National Park, Japan	7	36°11'01.7"	136°24'40.1"	Ju 189
	Mt. Ibuki, Kifu Prefecture, Japan	4	35°25'11.5"	121°36'32.04"	Ju 198
<i>F. longipetiolata</i>	Dujiangyan Longchi, Sichuan, China	10	31°06'09.46"	103°33'35.64"	Ju 587
	Badagong Mt. Hunan, China	4	29°27'42.21"	110°13'56.32"	Ju 648
<i>F. sylvatica</i>	Royal Botanic Gardens, Kew	5	51°28'26.25"	0°18'15.38"	Ju 1
	Epping Forest	5	54°40'13.53"	0°04'07.37"	Ju 533
	Windsor Great Park	5	51°25'57.59"	0°39'01.88"	Ju 538

Note: *N* = sample size.

APPENDIX 2. Information on voucher specimens for four *Fagus* species collected from arboretums or botanic gardens. All specimens are deposited at the Taiwan Forestry Research Institute (TAIF).

Species	Locality	<i>N</i>	Voucher no.
<i>F. grandifolia</i>	Royal Botanic Gardens, Kew	2	Ju 5
<i>F. lucida</i>	Royal Botanic Gardens, Kew	1	Ju 2
<i>F. orientalis</i>	Royal Botanic Gardens, Kew	1	Ju 4
<i>F. japonica</i>	Rokko Alpine Botanical Garden	4	Ju 408

Note: *N* = sample size.

APPENDIX 3. Primers for untested loci from *Fagus hayatae*.

Locus	Repeat motif	Allele size (bp)	Primer sequences (5'–3')
TB-6	(GAGAGT) ₈ N(GA) ₁₂	167	F: AGGTTTCGCACAAAACCCGTG R: AGTTTTTGGTTCGGCAAGGC
TB-32	(TC) ₁₅ (GA) ₁₅	190	F: GTAACCTCAATAACTAGATATTTAC R: TGAGTCCTGAGTAATCACAG
TB-38	(GGA) ₂ (GA) ₉ N(GGAG) ₄	164	F: GATGAGTCCTGAGTAAAGAC R: GACGTCAACAACCTGGGGAC
TB-42	(CT) ₃₄ (AG) ₁₆	159	F: TATAGATATATGGCCATGC R: GTCCTGAGTAATAAGCAACC
TB-47	(AT) ₅ (AG) ₁₄	167	F: AAGCTCCAGAAGAATCTTGC R: AAATTTCCCTGCACCACCT
TB-48	(CT) ₁₄ (AG) ₁₂	184	F: TGGCTGAAAAGTTCTACATG R: TCATTTGTTGCCACCACGT
TB-53	(CT) ₂₂	264	F: TCCATAAGGGTATAATGGTC R: TCTTGATGTGGAAAGTGTGC
TB-59	(AC) ₄₆	356	F: TTTACGCATGAATTTAGTACC R: ATCGATAAAGAGGAAGTCC
TB-93	(TG) ₃₉	285	F: CAGGCTCACTTTGCTGTTC R: GAGAGCGAAGATAGATTTTCC
TB-97	(CA) ₈ (AT) ₅ N(AC) ₃₂	225	F: GAATCAGATTGACCTATCC R: AAGTAGCCGTGACCCACGT
TB-103	(TGTA) ₅ N(GGTGAA) ₄	220	F: TGAACCTGAGATTTTGTG R: AAGTCTATTATGGTGTGC
TB-110	(CT) ₂₅ (AG) ₁₉	267	F: TTCGATTGATGAGTCTGAG R: CCCATTTCAACACCAATGCAC
TB-125	(GGA) ₁₄	323	F: ATGTGTGTAGGGGTGAGGTG R: CTAGCATTTCCAGCGGCA
TB-128	(TGT) ₁₁ (AGT) ₃ (TGT) ₉	248	F: GCTGGGATTACGGCTCCTGC R: AGGTGACATTTCGTACAGCA
TB-132	(GTT) ₁₄ (GA) ₁₈	228	F: TGAGCATTTTTCTTGTGCTGC R: ATTTTACCATTTCCTGCCA
TB-137	(AG) ₃₀	274	F: CGAAGTAGCAGGGGTTTGG R: ACCCATTTATGCCACCACCTTG
TB-139	(TG) ₃ (GTT) ₅ (GGT) ₃ (AGG) ₂	178	F: TTATTCTCCTTCTTCAAATTAGG R: GCTTCGACTCCTCGCTTCTC
TB-144	(GTT) ₁₆ (GAAGAT) ₃	248	F: GTCAAACTCAAACTCCCTTC R: TCTCTCATGTGCTACCTCG
TB-146	(GTT) ₁₀	311	F: AAGGGGTTGTGAGTGGTGG R: ACTCCATATGTTTACTAAGG
TBN-28	(CCATCA) ₁₀	221	F: TCCTGAGTAATTAGACTGCCG R: TACACTCGCCAAATGGGGGT
TBN-41	(AC) ₄₀	134	F: CAACAATATCATGAATTGAAAAT R: TGTGGATTCTCAGAGGCTC
TBN-45	(GA) ₁₈	192	F: GGACAATGGGGAGGAATGG R: ACTTTATTTCCCTTCCCCCT
TBN-47	(TCC) ₁₅	265	F: ACTCGAACTCCAACACCTG R: ACGAGGAAATCGAACCTGGTG
TBN-67	(GA) ₁₈	198	F: TAGCAAACCCACATCGACC R: ACTCAAAAAGACCAGTGCCA
TBN-110	(CT) ₂₅ (AG) ₃₄	392	F: TCAAACTTTTGTGTGCTGAGA R: GTTTTTGGCTCGGCAAAAGC
TBN-117	(AG) ₁₅	188	F: TTCTGTCCGGTTGGGGTAC R: ACTATATCTCCCATCCCTTCG